Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B. Tech CE 7th semester
Name of the Paper: ***Traffic Engineering and Management***
Paper Code: ***TCE 713***
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks) CO1

a. Explain the significance and scope of traffic engineering in modern transportation systems. How does it contribute to the efficiency and safety of road networks?

OR

b. Discuss the various characteristics of vehicles and road users that traffic engineers must consider when designing road infrastructure. How do these characteristics impact traffic flow and safety?

Q2. (10 Marks) CO1 & 2

a. Define skid resistance and braking efficiency. What factors affect these properties, and how can traffic engineers address issues related to skid resistance and braking in road design? Provide examples of problems that may arise due to inadequate skid resistance.

OR

b. What are the key components of traffic engineering? Discuss how road traffic and land use characteristics influence the design and planning of transportation systems.

Q3. (10 Marks) CO1 &2

a. Analyze the relationship between road traffic characteristics and land use patterns. How does effective traffic engineering balance these elements to improve mobility and reduce congestion in urban areas?

OR

b. What are the key types of traffic surveys used in transportation engineering? Discuss the methods for measuring traffic volume, capacity, and speed, and explain how this data is utilized in traffic planning and management.

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Q4. (10 Marks) CO2

a. Explain the process and significance of conducting origin and destination (O&D) surveys. How do these surveys help in understanding travel patterns, and what role do they play in the development of effective transportation strategies?

OR

b. Describe the methodologies used for analyzing parking and pedestrian studies in urban areas. How do these studies impact traffic management?

Q5. (10 Marks) CO2

a. Discuss the role of accident studies in traffic engineering. How are accident data collected.

OR

b. Define "level of service" (LOS) in traffic flow analysis. What are the basic principles of traffic flow that influence LOS, and how do delays, capacity, and speed affect the overall efficiency of road networks?